

Rotman

INTRO TO R PROGRAMMING

R Tutorial (RSM358) – Session 3

September 23, 2026 Prepared by Jay Cao / [MDAL](#)

Website: <https://rmdal.github.io/r-intro-2026-fall/>



Rotman School of Management
UNIVERSITY OF TORONTO

Any Questions about Lab 3.6?

- Loading the data provided by the textbook
 - Easy option: via the ISLR2 package
 - Alternatively: csv files on the textbook resource page (some are in the zipped file)
- Good practise: Before loading a csv data file, inspect it as a text file
 - This helps to determine the potential arguments of `read.csv()`
 - E.g., `header = FALSE` or `na.strings = "?"` or `stringsAsFactors = T`
- factor column/variable (categorical variable)
 - Why the factor data type?
 - Convert a non-factor column to a factor column
 - E.g. `auto$origin = as.factor(auto$origin)`

Any Questions about Lab 3.6?

- Linear regression?
- Comment on the regression summary report (section 3.4)
 - Is there a relationship, how strong, positive or negative
 - Confidence and prediction interval
- Interpretation of coefficients
 - Section 3.1 and 3.2 for quantitative/continuous predictors
 - Section 3.3.1 for qualitative/categorical predictors
 - Section 3.3.2 for interactive terms
- Outlier and high leverage observations
 - Use post-regression diagnostic plot; section 3.3.3

Lab 3.6 Linear Regression - 1

- Run a regression: `my_lm <- lm(formula = ..., data = ...)`
 - e.g. `lm(formula = y ~ x1, data = my_df)`
- Produce the regression report: `summary(my_lm)`
 - Know how to read test statistics
 - Know how to interpret the estimated coefficient;
- Confidence interval for the coefficients: `confint(my_lm)`

Lab 3.6 Linear Regression - 2

- `plot()`
 - Two variable scatter plot: `plot(x, y)`
 - Regression line: `abline(my_lm)`
 - Post-regression diagnostic plot: `plot(my_lm)`
- `predict(object, new_data, interval, level=0.95)`
 - Confidence interval of $\mathbb{E}(Y)$ at $X = x_0$
`predict(my_lm, data.frame(x1 = (c(5, 10))), interval = "confidence")`
 - Prediction interval of Y at $X = x_0$
`predict(my_lm, data.frame(x1 = (c(5, 10))), interval = "prediction")`

Create Data Frame using `data.frame()`

```
# create a data frame  
df1 <- data.frame(  
  x = 1:3,  
  y = letters[1:3],  
  z = c(1.1, 2.2, 3.3)  
)
```

x	y	z
1	"a"	1.1
2	"b"	2.2
3	"c"	3.3

Understanding CI and PI - Setup

- True model, true Y , and true coefficients β_i ,

$$Y = f(X) + \epsilon = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon, \text{ where } \mathbb{E}(\epsilon) = 0.$$

- Estimated model, predicted $\hat{Y}$, and estimated coefficients $\hat{\beta}_i$,

$$\hat{Y} = \hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2$$

- R code

```
my_lm <- lm(formula = y ~ x1 + x2, data = my_df)
```

CI of β_i

- Under “usual/standard” assumptions,

$$\frac{\hat{\beta}_i - \beta_i}{\widehat{SE}(\hat{\beta}_i)} \sim t_{n-p},$$

So,

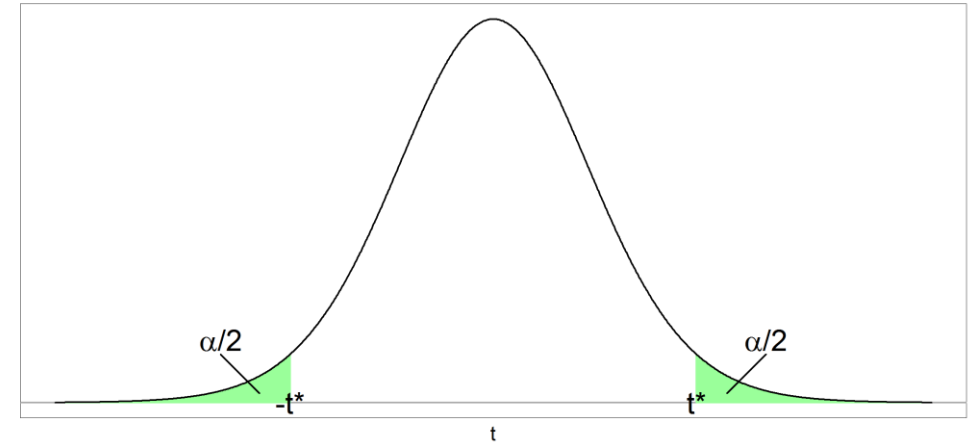
$$\mathbb{P}\left(-t_{n-p, \frac{\alpha}{2}} < \frac{\hat{\beta}_i - \beta_i}{\widehat{SE}(\hat{\beta}_i)} < t_{n-p, \frac{\alpha}{2}}\right) = 1 - \alpha,$$

Now, we get CI (centered around $\hat{\beta}_i$)

$$[\hat{\beta}_i - \widehat{SE}(\hat{\beta}_i)t_{n-p, \frac{\alpha}{2}}, \hat{\beta}_i + \widehat{SE}(\hat{\beta}_i)t_{n-p, \frac{\alpha}{2}}]$$

- Interpretation (e.g., $\alpha = 95\%$)?
 - “If we take repeated samples and construct the confidence interval for each sample, 95% of the intervals will contain the true unknown value of the parameter.” – from your textbook
 - That is, the 95% refers to the reliability of the method, not the single calculated range.
- R code: `confint(my_lm)`

Critical value t^* given α : two sided



Notations: t_{n-p} is t -dist w/ df $n - p$, n is # of obs., and p is # of parameters; α is significance level and $t_{n-p, \frac{\alpha}{2}}$ is critical value (the t^* in the graph) such that the prob of the t_{n-p} distribution to the right of it is $\frac{\alpha}{2}$; $\widehat{SE}(\hat{\beta}_i)$ is $SE(\hat{\beta}_i)$ in your textbook.

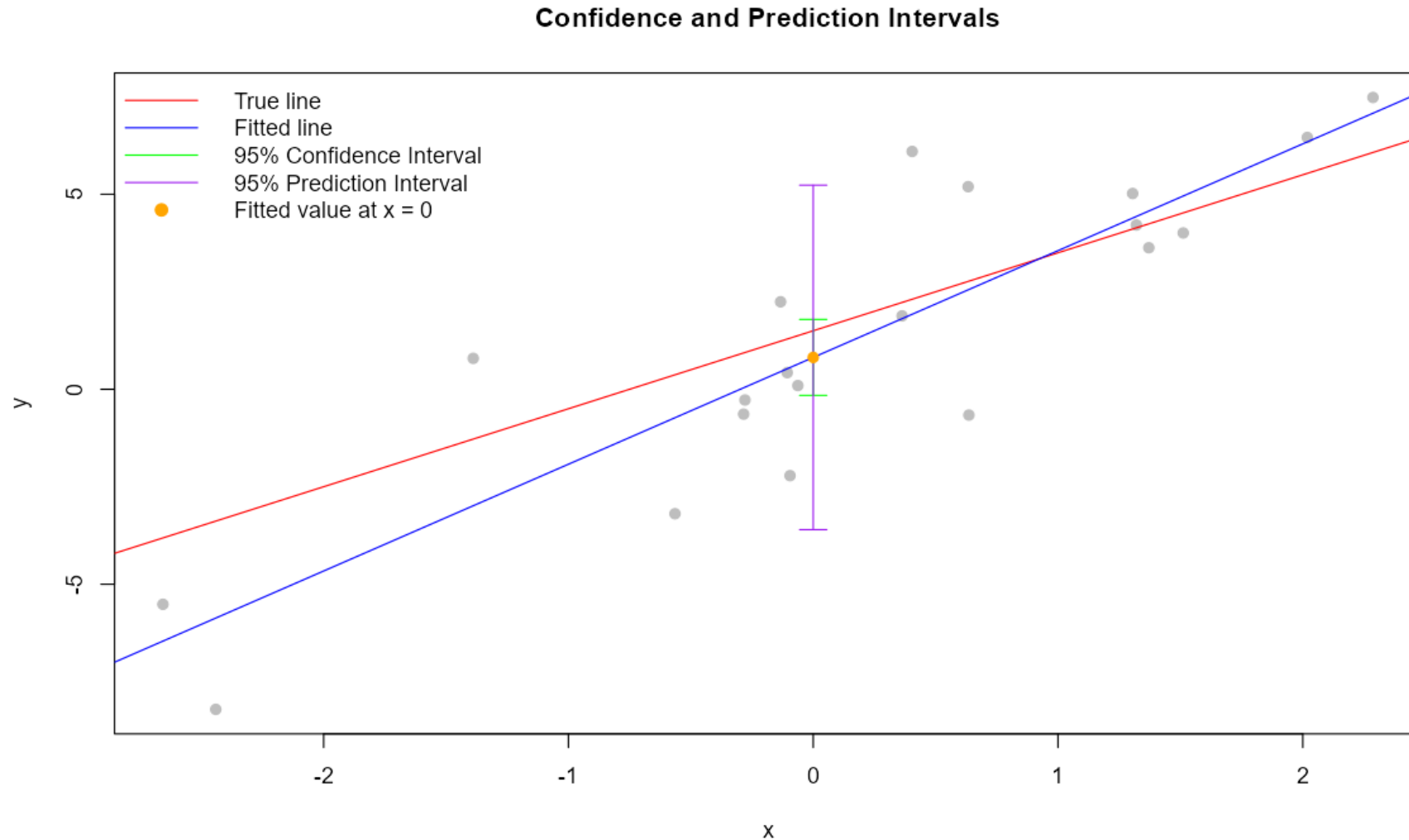
CI of $\mathbb{E}(Y) = f(X)$

- α level CI of $\mathbb{E}(Y) = f(X)$ at $X = x_0$ can be derived similarly
 - It's an interval centered at $\hat{f}(X = x_0)$
- Interpretation (say, $\alpha = 95\%$): “95% of intervals **of this form** will contain the true value of $f(X)$ ” – from your textbook
 - To be precise, $f(X = x_0)$
 - What does it mean “of this form”?
 - If we take repeated samples and construct the confidence interval for each sample, ...
- R code
 - `predict(my_lm, new_data, interval = "confidence")`

PI of $Y = f(X) + \epsilon$

- α level PI of $Y = f(X) + \epsilon$ at $X = x_0$ can be derived similarly
 - Again, it's an interval centered at $\hat{f}(X = x_0)$
- Wider than the corresponding CI
 - because it accounts for the irreducible error
- Interpretation?
- R code
 - `predict(my_lm, new_data, interval = "prediction")`

CI of $\mathbb{E}(Y)$ and PI of Y at $X = 0$ and $\alpha = 95\%$



Note: Data are simulated with the true model being $Y = 1.5 + 2X + \epsilon$ where $\epsilon \sim N(0,2)$.

lm() R Regression Formula - 1

my_df

y	x1	x2	x3
18	8	307	130
16	8	304	150
...	...	...	...

lm() Regression Formula	Regression Formula
<code>lm(formula = y ~ x1 + x2 + x3, data = my_df)</code> <code>lm(formula = y ~ ., data = my_df)</code>	$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon$
<code>lm(formula = y ~ . - x3, data = my_df)</code> <code>lm(formula = y ~ x1 + x2, data = my_df)</code>	$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$
<code>lm(formula = y ~ 0 + x1 + x2 + x3, data = my_df)</code>	$Y = \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon$
...	...

Source: <https://stat.ethz.ch/R-manual/R-devel/library/stats/html/formula.html>

lm() R Regression Formula - 2

my_df

y	x1	x2	x3
18	8	307	130
16	8	304	150
...	...	...	...

lm() Regression Formula	Regression Formula
<code>lm(formula = y ~ x1 * x2, data = my_df)</code>	$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \epsilon$
<code>lm(formula = y ~ x1 + x2 + x1:x2, data = my_df)</code>	
<code>lm(formula = y ~ x1 + x2 + I(x1 * x2), data = my_df)</code>	
<code>lm(formula = y ~ x1 + I(x1^2), data = my_df)</code>	$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \epsilon$
<code>lm(formula = y ~ x1 + log(x2), data = my_df)</code>	$Y = \beta_0 + \beta_1 X_1 + \beta_2 \ln(X_2) + \epsilon$
...	...

Source: <https://stat.ethz.ch/R-manual/R-devel/library/stats/html/formula.html>

Lab 4.7 Logistic Regression

- `my_model <- glm(formula = ..., data = ..., family=binomial)`
- `summary(my_model)`
- `predict(my_model, newdata = ..., type = "response")`
 - Set the argument `type = "response"` to get predicted probabilities, i.e., $P(Y = 1|X)$
 - Otherwise, `predict(my_model)` gives log odds (logit)
 - If the `newdata` argument is not supplied, the prediction is applied on the training data set
 - Use `contrast()` to find out which y category is set to 1.
- Construct confusing matrix
 - Convert probability prediction to binary prediction (cutoff prob.)
 - `table()`

Lab 4.7 Training & Test Set

- Training and test set split
 - For time series data, need to respect the time when splitting the data
 - That is, train on early data, test on late data
 - Otherwise, randomly split data to train and test

```
# randomly split Auto dataset into training and test set
num_rows <- nrow(Auto)
train_fraction <- 0.7
train_idx = sample(1:num_rows, size = round(num_rows * train_fraction))
train_data <- Auto[train_idx, ]
test_data <- Auto[-train_idx, ]
```